Chapter 1: Defining and Collecting Data

Basic Business Statistics: Concepts and Applications, 13th Edition — Comprehensive Chapter Summary, Business Applications, and Teaching Guide

This guide consolidates everything we have discussed about Chapter 1, preserving the original summary and expanding it with business cases, practical examples, classroom questions, sampling-method comparisons, and explanations of survey errors.

The objective is not merely to teach students statistical terminology. It is to help them understand why statistics matters in business, how it supports decisions, and how poor data can lead to costly mistakes.

Part I — Understanding the Chapter

1. Chapter overview: Why data matters

The central message of Chapter 1 is:

Good statistical analysis begins with good data.

Even the most sophisticated statistical techniques cannot produce trustworthy conclusions if:

- Variables are poorly defined.
- Data are inaccurate or incomplete.
- Samples are biased or unrepresentative.
- Data collection methods are inappropriate.
- Survey questions are misleading.

This chapter addresses five major topics:

1. Defining data and variables.
2. Understanding measurement scales.
3. Collecting and preparing data.
4. Selecting appropriate sampling methods.
5. Identifying survey errors and ethical concerns.

These are foundational concepts for statistical analysis, business analytics, market research, financial analysis, quality management, and evidence-based decision-making.

Why should business students care?

Consider a company planning to invest $20 million in a new product.

Management might want to know:

- How large is the potential market?
- What percentage of customers would purchase the product?
- What price are customers willing to pay?
- How does demand vary across customer segments?

- How reliable are the survey findings?

Before calculating an average, building a regression model, or testing a hypothesis, the company must establish what information is needed and how to obtain it.

Statistics begins with understanding the business problem, not selecting a statistical test.

2. The DCOVA framework

The textbook introduces the DCOVA framework as a structured approach to applying statistics to business decisions.

D

Define

C

Collect

O

Organize

V

Visualize

A

Analyze

Chapter 1 concentrates on Define and Collect; later chapters develop the remaining tasks.

Stage	Main question	Example
Define	What should we measure?	Customer satisfaction, sales, profitability
Collect	Where will we obtain the data?	Surveys, transactions, company records
Organize	How should we structure the data?	Tables and frequency distributions
Visualize	How should we present the data?	Charts and graphs
Analyze	What conclusions can we draw?	Trends, relationships, statistical inference

The textbook emphasizes that establishing the business objective comes before applying DCOVA.

Business example

A retail company notices declining profits.

It could immediately begin analyzing its sales database. But what if the problem is not declining sales?

Perhaps sales are increasing while operating costs rise even faster.

The business objective must therefore be defined carefully.

A more useful question might be:

What factors explain the decline in operating profit during the past three years?

Only then should analysts identify the relevant variables and begin collecting data.

Part II — Defining Data and Variables

3. Defining variables (Section 1.1)

A variable is a characteristic that can take different values across observations.

Examples include:

- Customer age.
- Monthly income.
- Number of purchases.
- Product category.
- Customer satisfaction.
- Delivery time.

However, simply naming a variable is insufficient.

Operational definitions

An operational definition specifies exactly what a variable means and how its values are determined.

Suppose a manager requests data on annual sales.

That raises several questions:

- Does annual sales mean revenue or number of units sold?
- Does it cover the calendar year or fiscal year?
- Does it include returns and refunds?
- Does it include online and physical stores?
- Does it represent one branch or the entire company?

Without operational definitions, different analysts might produce different results while believing they are measuring the same thing.

Classroom example

Ask students:

> A CEO requests a report on employee productivity. What exactly should the analysts measure?

Possible answers include:

- Units produced per employee per hour.
- Revenue generated per employee.
- Number of customer cases resolved.
- Average time required to complete a task.

These measures are not interchangeable.

The lesson is that a business concept must be translated into a measurable variable before statistical analysis can begin.

4. Types of variables

Variables fall into two major categories.

Variables

Categorical (Qualitative)

Values represent groups or labels.

Examples: payment method, product type, academic major.

Numerical (Quantitative)

Discrete: arise from counting, such as number of purchases.

Continuous: arise from measurement, such as delivery time or weight.

Discrete versus continuous

A discrete variable generally takes countable numerical values.

For example, the number of customers entering a store could be 100 or 101, but not 100.5 customers.

A continuous variable represents a measured quantity that can theoretically take any value within an interval.

For example, a delivery might take 12 minutes, 12.5 minutes, or 12.53 minutes, depending on measurement precision.

An important subtlety from the textbook is that a variable's type depends on how it is defined.

Age can be numerical when recorded in years, but categorical when classified into groups such as children, young adults, middle-aged adults, and seniors.

Business significance

The variable type influences which statistical methods and visualizations are appropriate.

For example, calculating the average of nominal product categories is meaningless, while calculating the average delivery time is meaningful.

Part III — Measurement Scales

5. The four measurement scales (Section 1.2)

The chapter introduces four scales.

Scale	Description	Examples
Nominal	Categories without ranking	Payment method, service provider
Ordinal	Ordered categories	Satisfaction levels, product ratings
Interval	Numerical differences are meaningful, but zero is arbitrary	Celsius temperature
Ratio	Numerical differences and ratios are meaningful; zero is genuine	Sales revenue, age, weight

Nominal scale

Nominal data classify observations into categories without implying order.

For example, customers may use cash, credit cards, debit cards, or digital wallets.

These categories are different, but one is not inherently greater than another.

Ordinal scale

Ordinal data contain categories that can be ranked.

For example:

Poor → Fair → Good → Very Good → Excellent

The ranking is meaningful, but the distance between categories is not necessarily equal.

The difference between Poor and Fair need not equal the difference between Good and Very Good.

Interval scale

Interval data have meaningful differences between values, but lack a true zero point.

For example, 30°C is 10 degrees warmer than 20°C.

However, 30°C is not 1.5 times as hot as 20°C on the Celsius scale.

The textbook also gives standardized examination scores as interval-scale examples.

Ratio scale

Ratio data have a meaningful zero, so both differences and ratios can be interpreted.

For example:

- $1,000 is twice $500.
- A 20-minute delivery takes twice as long as a 10-minute delivery.

- A product weighing 4 kilograms is twice as heavy as one weighing 2 kilograms.

The chapter notes that temperature measured in Kelvin has a true zero and is therefore ratio-scaled.

Why this matters

Measurement scales determine what numerical operations and statistical comparisons are meaningful.

For a master's class, it is also useful to explain that the appropriate analytical method depends on more than the scale alone. Research design, distributional assumptions, sample structure, and the objective of the analysis also matter.

Part IV — Collecting and Preparing Data

6. Data collection (Section 1.3)

Once variables have been defined, analysts must determine how to collect the required data.

The textbook distinguishes between primary and secondary data.

Primary data	Secondary data
Collected directly for the analysis	Previously collected by another party
Customer survey conducted by the researcher	Published market research
Company experiment	Government economic statistics
Direct observation	Third-party industry database

The distinction depends on the role of the researcher and the purpose of collection.

The chapter identifies five broad sources of data:

1. Data distributed by organizations or individuals.
2. Designed experiments.
3. Surveys.
4. Observational studies.
5. Data generated by ongoing business activities.

Experiments versus surveys versus observation

These approaches answer different questions.

Experiment: A company randomly assigns customers to two advertising versions and compares purchasing behavior. This can help establish causal effects when properly designed.

Survey: A company asks customers which advertisement they prefer. This measures reported preferences rather than necessarily demonstrating actual purchasing behavior.

Observational study: A retailer studies how customers move through a store without assigning experimental conditions.

Operational data: The retailer analyzes transactions from its point-of-sale system.

These methods can complement one another, but their findings do not necessarily have the same interpretation.

7. Population versus sample

A population contains all individuals or items about which researchers want to draw conclusions.

A sample contains a selected portion of that population.

Suppose a university has 10,000 students and surveys 500.

- Population: 10,000 students.
- Sample: 500 selected students.
- Population parameter: the true proportion of all students who are satisfied.
- Sample statistic: the proportion of the 500 respondents who report satisfaction.

The sample statistic can be used to estimate the population parameter when the sampling design and assumptions support that inference.

Why sample instead of surveying everyone?

The textbook identifies three practical reasons:

- Lower cost.
- Less time.
- Easier administration and analysis.

Sampling is especially important when populations are very large or geographically dispersed.

A census examines the entire population and therefore does not involve sampling error, but it can still suffer from coverage, nonresponse, and measurement problems.

8. Structured and unstructured data

Structured data follow a predefined organization, such as rows and columns.

Examples include:

- Sales transaction tables.
- Customer databases.
- Payroll records.
- Inventory systems.

Unstructured data lack a consistent tabular format.

Examples include:

- Customer emails.

- Social media posts.
- Open-ended survey responses.
- Free-form business documents.

The textbook also recognizes semistructured data, such as forms that combine fixed fields with open-ended responses.

This distinction is particularly relevant to modern business analytics.

A retailer may analyze structured purchase histories alongside unstructured customer reviews to understand both what customers buy and what they think about those products.

The chapter also discusses electronic formats and encodings. Some formats are immediately suitable for statistical analysis, while others require conversion. Numerical encodings can affect precision and compatibility across data sources.

9. Data cleaning

Before analysis, researchers must identify data irregularities.

Examples include:

Problem	Example
Missing value	Customer does not report income
Impossible value	Customer age recorded as 250
Undefined category	Payment method recorded outside permitted categories
Outlier	An unusually large transaction
Inconsistent format	Dates recorded using incompatible formats

An outlier is not automatically an error.

A transaction of $100,000 may be suspicious in a dataset of ordinary retail purchases, but it could represent a legitimate corporate order.

The analyst must investigate before deciding how to handle it.

Recoding variables

Recoding means transforming existing values into new categories or variables.

For example, numerical age might be grouped into:

- Under 18.
- 18–34.
- 35–54.
- 55 and above.

Two principles are essential.

Mutually exclusive: Every observation belongs to only one category.

Collectively exhaustive: Every observation can be assigned to a category.

For example, age categories of 18–30 and 30–45 are problematic because someone aged 30 belongs to both.

The textbook also illustrates recoding students from four academic classifications into lowerclassmen and upperclassmen.

Part V — Probability Sampling Methods

10. Sampling fundamentals (Section 1.4)

Before selecting a sample, researchers need a sampling frame: a complete or partial list of the population members from which selection will occur.

For example, a university's enrollment database could serve as the sampling frame for a student survey.

If the database excludes some eligible students, the sample may suffer from coverage error.

The chapter distinguishes two major sampling approaches.

Sampling

Nonprobability

Selection probabilities are unknown.

Convenience sampling

Judgment sampling

Probability

Selection is based on known probabilities.

Simple random

Systematic

Stratified

Cluster

Probability sampling provides the conventional foundation for design-based statistical inference.

Nonprobability sampling is generally faster and cheaper but creates difficulties when generalizing findings to a wider population.

Nonprobability sampling

Convenience sampling: Select participants who are easy to reach.

Example: A professor surveys students attending today's lecture.

Judgment sampling: Select participants based on expertise or researcher judgment.

Example: A company interviews ten senior cybersecurity specialists about emerging threats.

Both approaches may be useful for exploratory research, pilot studies, or expert insights, but they do not automatically provide representative population estimates.

11. Simple random sampling

Randomly selected individuals are shown in blue.

In simple random sampling, every possible sample of a given size has the same probability of selection.

Example: A company has 2,000 employees and randomly selects 200 for a satisfaction survey.

When to use it

Use simple random sampling when:

- A complete sampling frame is available.
- The population is reasonably accessible.
- There is no need to guarantee representation of particular subgroups.
- A straightforward random selection process is practical.

Advantages

It is conceptually simple and provides a strong foundation for standard statistical inference.

Limitations

It can be expensive for geographically dispersed populations and may produce relatively few observations from small subgroups.

Sampling with and without replacement

The textbook distinguishes two selection procedures.

With replacement: A selected individual can be selected again.

Without replacement: Once selected, an individual cannot be selected again.

Most employee and customer surveys use sampling without replacement because repeatedly selecting the same person is generally undesirable.

12. Systematic sampling

Every fourth item is selected after a random start.

Systematic sampling selects every \\(k\\)th item after choosing a random starting position.

The interval is:

\\[k=\frac{N}{n} \\]

where \\(N\\) is the frame size and \\(n\\) is the desired sample size.

Example

A factory produces 10,000 items daily and wants to inspect 200.

\\[k=\frac{10,000}{200}=50 \\]

Select a random starting number between 1 and 50, then inspect every fiftieth item.

When to use it

Systematic sampling is particularly suitable for:

- Production lines.
- Sales invoices.
- Customer transaction records.
- Membership directories.
- Ordered administrative lists.

Advantage

It is fast and easy to implement.

Limitation

It can produce biased results when the ordering of the population contains periodic patterns associated with the variable being measured.

For example, if every fiftieth manufactured product is processed by a different machine, selecting every fiftieth item could distort the quality estimate.

13. Stratified sampling

Randomly selected individuals from every stratum are highlighted.

Stratified sampling divides the population into meaningful subgroups, called strata, and randomly samples individuals from each subgroup.

Example: University survey

A university has 10,000 students:

Stratum	Population	Proportionate sample
Undergraduate	6,000	300
Master's	3,000	150

| PhD | 1,000 | 50 |

| Total | 10,000 | 500 |

Every academic level contributes to the sample.

When to use it

Use stratified sampling when:

- Important subgroups must be represented.
- Researchers want separate estimates for those subgroups.
- Subgroups differ meaningfully in the characteristic being studied.
- Stratification can improve statistical precision.

Advantages

Stratification can reduce sampling variability and ensure that small but important groups are included.

Limitations

It requires information about subgroup membership and can be more complicated to implement.

Proportionate versus disproportionate stratification

In proportionate stratification, sample sizes follow population proportions.

In disproportionate stratification, some groups receive larger or smaller samples relative to their population sizes.

For example, a university might deliberately sample 150 PhD students rather than 50 because it needs a more precise estimate of PhD satisfaction.

When estimating the overall student population, appropriate sampling weights may be necessary.

14. Cluster sampling

Entire branches are selected as clusters.

Cluster sampling divides the population into naturally occurring groups, randomly selects some groups, and studies the individuals in those selected groups.

Example: National bank

A bank operates 100 branches across the country.

Surveying customers at every branch would be expensive.

Instead, researchers randomly select 10 branches and collect data from all eligible customers in those selected branches.

When to use it

Cluster sampling is particularly useful when:

- The population is geographically dispersed.
- Natural groups already exist.
- Travel and administration costs are substantial.
- A complete list of individuals is difficult to obtain, but a list of groups is available.

Advantages

It can substantially reduce fieldwork and travel costs.

Limitations

Individuals within a cluster may resemble one another, reducing statistical precision.

For example, customers of a branch located in an affluent neighborhood may have similar incomes and spending patterns.

A sample consisting of a small number of such branches may not represent the national customer base well.

Two-stage cluster sampling

A common extension is to:

1. Randomly select branches.
2. Randomly select customers within those branches.

This is known as two-stage cluster sampling.

The textbook's basic illustration focuses on studying all items within selected clusters.

15. Stratified versus cluster sampling: The critical distinction

This is perhaps the most important conceptual comparison in the sampling section.

Both methods divide a population into groups, but they use those groups differently.

Stratified

Representation and precision

Sample from every group.

Divide employees by department and randomly select employees from each department.

Cluster

Cost and practicality

Select only some groups.

Divide employees by branch and randomly select entire branches.

A further distinction concerns group composition.

Stratified sampling: We generally want members within a stratum to be relatively homogeneous regarding the variable of interest.

Cluster sampling: Ideally, each cluster resembles a miniature version of the population, containing a mixture of different types of individuals.

These are design principles rather than absolute requirements.

Comparing all four probability methods

Method	How selection works	Best use
Simple random	Random individuals from the entire frame	Complete population list
Systematic	Every \\(k\\)th item after a random start	Ordered lists and production lines
Stratified	Random individuals from every subgroup	Subgroup representation and precision
Cluster	Randomly selected groups	Geographic dispersion and cost reduction

A practical decision rule

- Simple random: Can we randomly select individuals directly?
- Systematic: Can we efficiently select every \\(k\\)th item?
- Stratified: Do we need representation from every important subgroup?
- Cluster: Can we reduce costs by selecting only some groups or locations?

The easiest way to remember the distinction is:

Stratified sampling selects individuals from all groups. Cluster sampling selects some groups.

Part VI — Survey Errors

16. The four survey errors (Section 1.5)

The textbook identifies four major sources of survey error.

Error	Definition	Typical consequence
Coverage	Some population members are excluded from the frame	Selection bias
Nonresponse	Selected individuals fail to provide data	Potential nonresponse bias
Sampling	Random differences between samples	Sampling uncertainty
Measurement	Collected values do not accurately measure the intended characteristic	Inaccurate observations

17. Coverage error

Coverage error occurs when the sampling frame fails to properly represent the target population.

Examples

Banking: A bank surveys only mobile banking users when it wants to understand all customers. Branch-only customers are excluded.

University: A university surveys students using a list containing only online-course enrollments. Students taking exclusively face-to-face courses are missing.

Retail: A supermarket surveys customers using only its loyalty-program database. Customers without loyalty cards cannot be selected.

The important point is that excluded individuals never had a chance to participate.

Business consequence

Management may make decisions that favor the customers included in the frame while overlooking other customer segments.

18. Nonresponse error

Nonresponse error arises when selected individuals fail to provide the requested data.

It can produce bias when nonrespondents differ systematically from respondents.

Examples

Employee satisfaction: A company randomly selects 500 employees, but only 200 respond. Employees afraid of retaliation may be less willing to participate.

Banking: A bank selects 1,000 customers for a satisfaction survey. If dissatisfied customers respond at lower rates, the findings may overestimate satisfaction.

University: Students facing serious academic difficulties may be less likely to complete a student experience survey.

The problem is not merely that some responses are missing.

The central question is whether the missing responses differ systematically from the responses collected.

How can researchers reduce nonresponse problems?

The textbook discusses:

- Follow-up attempts.
- Incentives.
- Alternative survey modes.
- Comparing early and later responses.
- Improving response rates through better survey administration.

A high response rate is desirable, but it does not automatically guarantee the absence of nonresponse bias.

19. Sampling error

Sampling error arises because a sample contains only part of the population.

Even when a probability sample is selected correctly, sample estimates can differ from the true population values.

Examples

Retail: The true satisfaction rate is 70%. One random sample estimates 67%, while another estimates 73%.

Political polling: Two independently selected random samples estimate candidate support at 48% and 51%.

Manufacturing: The actual average product weight is 500 grams. One sample estimates 498 grams, another 502 grams.

These differences can occur without mistakes in data collection.

Can sampling error be reduced?

Generally, increasing sample size reduces sampling variability, all else equal.

For many simple random sampling estimates, standard error decreases approximately in proportion to:

\\[\frac{1}{\sqrt{n}} \\]

For example, quadrupling the sample size approximately halves the standard error under the usual assumptions.

However, increasing sample size does not automatically remove coverage, nonresponse, or measurement bias.

This is one of the most important lessons in the chapter.

20. Measurement error

Measurement error occurs when collected data do not accurately represent the characteristic being measured.

Examples

Leading questions: Instead of asking, "How satisfied are you with our service?", a survey asks, "How satisfied are you with our excellent service?"

Income reporting: Respondents underreport their income because they consider it sensitive.

Employee performance: Two managers rate equivalent performance differently because evaluation criteria are vague.

Customer spending: Customers are asked to remember their total spending over the previous year but cannot recall it accurately.

Common causes

The chapter identifies several sources, including:

- Leading questions.
- Ambiguous wording.
- Double-barreled questions that ask about two issues at once.
- Poorly defined reference periods.
- Social desirability bias.
- Memory limitations.
- Interviewer influence.
- Respondents providing inaccurate information.

Classroom example

Compare these two questions:

Question A: How satisfied are you with our excellent customer service?

Question B: Overall, how satisfied or dissatisfied are you with the customer service you received during your most recent visit?

Question B is more neutral and specifies the experience being evaluated.

21. Can all four errors occur in one survey?

Yes.

Suppose a university wants to measure student satisfaction.

It randomly selects 1,000 students from a database containing only full-time students.

Only 400 respond.

The questionnaire asks:

How satisfied are you with our outstanding teaching quality?

Problem	Error
Part-time students excluded from a survey intended for all students	Coverage
Only 400 of 1,000 selected students respond	Potential nonresponse
Random variation from selecting 1,000 students	Sampling
Leading question wording	Measurement

This is an excellent classroom exercise because it shows that the errors are not mutually exclusive.

A survey can suffer from several problems simultaneously.

22. Ethical issues in surveys

The chapter explains that survey errors can also involve ethical concerns.

Examples include:

- Deliberately excluding certain groups from a sampling frame.
- Designing surveys so that particular groups are less likely to respond.
- Hiding sample sizes or margins of error.
- Using leading questions to manipulate responses.
- Presenting nonprobability survey findings as though they represent an entire population.
- Knowingly providing false survey responses.

For business students, the key message is that statistical integrity is not simply a technical requirement. It is also a professional and ethical responsibility.

Part VII — The Literary Digest Case

23. Why a huge sample produced the wrong prediction

The chapter presents the famous 1936 U.S. presidential election poll conducted by Literary Digest.

Ballots distributed

10 million

Responses received

2.3+ million

The magazine predicted that Alf Landon would defeat Franklin D. Roosevelt.

Instead, Roosevelt won decisively.

The case demonstrates that a large sample does not guarantee accuracy.

The textbook notes that the failure was traditionally attributed to coverage bias because the sampling frame disproportionately reached wealthier individuals.

However, later analyses also highlighted low response rates and differential nonresponse as important explanations.

Teaching discussion

Ask students:

> Would you trust a survey with two million respondents more than one with 2,000 respondents?

Many students may initially say yes.

Then introduce the distinction between sample size and sample representativeness.

A properly designed probability sample of 2,000 individuals can provide a more reliable estimate than two million self-selected responses.

The same problem remains relevant to:

- Online opinion polls.
- Social media surveys.
- Website feedback forms.
- Customer satisfaction surveys.
- Voluntary product-review systems.

The lesson is:

A very large biased sample can be less reliable than a much smaller, properly selected probability sample.

Part VIII — Business Cases from the Chapter

This section is particularly useful for demonstrating the practical value of statistics to master's students.

The chapter includes several business and educational cases. Some provide detailed business narratives, while others are exercises that require additional downloadable datasets.

24. Case 1: Good Tunes & More and Whitney Wireless

Textbook pages 41, 55–56 and 59–60

Business background

Good Tunes & More originally sold music CDs and vinyl records.

As consumer preferences changed, the company expanded into home entertainment and computer systems.

Despite expansion, GT&M experienced:

- Disappointing revenue growth.
- Declining profit margins.
- Increasing competition from online retailers.
- Uncertainty about its future business model.

Some investors recommended closing stores and reducing merchandise variety.

CEO Emma Levia instead decided to pursue expansion by acquiring Whitney Wireless, a successful three-store retailer specializing in smartphones and mobile devices.

Her vision was to create an electronics retailer offering a comprehensive range of products.

The business decision

How much should GT&M pay to acquire Whitney Wireless?

The analysts must collect data that will help management determine a fair acquisition price.

The chapter emphasizes the need to define variables carefully and identify appropriate data sources.

Data identified in the chapter

The analysts focus on financial variables associated with the company's profit and loss statement, particularly sales and expenses.

They also identify external economic information and customer demographic data from the companies' loyalty programs.

Social media data may provide supplementary information about potential customers.

Suggested classroom extension

Ask students to identify additional variables that would help evaluate the acquisition.

Variable	Why it matters
Annual revenue	Measures sales performance
Revenue growth	Indicates business expansion or decline
Gross profit margin	Evaluates profitability of sales
Operating expenses	Measures operating efficiency
Net profit	Measures earnings after expenses
Debt obligations	Helps assess financial risk
Customer retention	Indicates loyalty and repeat business
Sales by store	Identifies strong and weak locations
Market growth	Helps evaluate future prospects

These additional variables are teaching suggestions, not a complete list explicitly provided by the textbook.

Classroom discussion questions

1. Is increasing sales revenue sufficient evidence that a company is becoming more valuable?
2. Could a company increase sales while reducing profitability?
3. Which data should be collected from Whitney Wireless, and which should come from external sources?
4. Would customer demographic data help determine whether the acquisition creates strategic value?
5. How would inconsistent definitions of revenue or profit affect the acquisition analysis?

Main lesson

Statistics helps management move from intuition and business claims toward evidence-based acquisition decisions.

However, the chapter stops at identifying and collecting data. It does not perform a financial valuation or calculate an acquisition price.

25. Case 2: Whitney Wireless — The best sales year ever?

Digital case discussion, textbook pages 59–60

This is a particularly useful case for teaching critical thinking.

Whitney Wireless claims to have achieved its best sales year ever.

But the supporting visualization has several problems.

The textbook identifies issues such as:

- Symbols without clearly defined numerical scales.
- Different symbol sizes that make comparisons ambiguous.
- Partial icons that are difficult to interpret.
- Unclear distinctions between unit sales and dollar sales.
- Ambiguous time labels.
- Incomplete historical comparisons.

For example, the chart compares sales from "two years ago" with sales from the "latest twelve months."

But these are not necessarily equivalent reporting periods.

Furthermore, a claim of the best sales year ever should be supported by comparisons with all relevant historical years, not merely two selected periods.

Classroom exercise

Tell students:

> Whitney Wireless reports its best sales year ever. Management wants you to verify the claim before GT&M acquires the company.

Ask them what evidence they would request.

Possible answers:

- Annual revenue for every year since establishment.
- Annual unit sales.
- Sales by product category.
- Gross and operating profit.
- Consistent reporting periods.
- Comparable accounting definitions.

Then ask:

Can a chart be visually impressive while statistically misleading?

Absolutely.

The case demonstrates that data visualization is not merely about making charts attractive. It is about communicating accurate, interpretable information.

26. Case 3: Ashland MultiComm Services

Textbook page 58

Business background

Ashland MultiComm Services (AMS) provides communications services, including:

- Digital cable television.
- Telephone services.
- High-speed internet access.

The company faces increasing competition from other network providers.

Management has observed declines in new digital cable installations and subscription renewals.

It also suspects that internet-based content delivery may be affecting the traditional cable television business.

The business problem

Management wants to understand why the business is losing momentum and what actions might improve performance.

Potential strategies include:

- Increasing promotional spending.
- Adjusting subscription fees.
- Improving customer service.

But management needs evidence before choosing a strategy.

Data mentioned in the chapter

The managers propose examining historical and year-to-date information on:

- Number of subscribers.
- Revenue.
- Subscription renewal rates.

The case asks students to identify other data sources and appropriate collection techniques.

Suggested classroom extension

Ask students what additional variables they would examine.

Variable	Business purpose
Monthly cancellations	Measure customer attrition
Customer complaints	Identify service problems
Subscription price	Assess pricing competitiveness

| Competitor prices | Compare market positioning |

| Internet streaming usage | Investigate substitution |

| Customer satisfaction | Assess perceived service quality |

| Promotional spending | Evaluate marketing investment |

| Renewal rate | Measure customer retention |

Classroom discussion

Suppose AMS reports that subscription renewals declined by 15%.

Ask:

> Is the decline caused by higher prices, poor customer service, or customers switching to streaming platforms?

The observed decline alone cannot establish the cause.

Students should recognize the need for additional variables, appropriate comparisons, and potentially further statistical analysis.

A later course could use this case for regression analysis, hypothesis testing, or time-series forecasting.

Main lesson

Statistics helps managers distinguish between symptoms and possible causes before committing resources to a business strategy.

27. Case 4: CardioGood Fitness

Textbook pages 58–59

Business background

CardioGood Fitness manufactures cardiovascular exercise equipment.

The company wants to increase treadmill sales and hires the AdRight advertising agency to identify customer segments and target advertising more effectively.

CardioGood sells three treadmill models.

| Model | Market position | Price in the case |

| ----- | ---------- | ---------- |

| TM195 | Entry-level treadmill | $1,500 |

| TM498 | Midrange treadmill | $1,750 |

| TM798 | Premium treadmill | $2,500 |

The agency wants to determine whether customers purchasing different models have distinct demographic and behavioral profiles.

Variables collected

The chapter identifies the following variables:

Variable	Classification
Treadmill model	Categorical, nominal
Gender	Categorical, nominal
Age	Numerical
Education in years	Numerical
Relationship status	Categorical, nominal
Annual household income	Numerical
Expected weekly treadmill usage frequency	Numerical, discrete when counted as sessions
Expected weekly distance	Numerical, continuous
Self-rated fitness, 1–5	Categorical, ordinal

The classification of some numerical variables depends on their operational definitions and recording precision.

The business questions

- Do higher-income customers tend to purchase more expensive treadmills?
- Are premium treadmill buyers generally more experienced exercisers?
- Do customers of different ages prefer different models?
- Is expected weekly usage associated with the product selected?
- Can the company identify customer segments for targeted advertising?

Suggested classroom analysis

Students could begin with descriptive statistics:

- Average age by treadmill model.
- Average income by model.
- Frequency distribution of fitness ratings.
- Percentage of customers purchasing each model.

Later, the same case could support:

- ANOVA to compare mean income across product groups.
- Chi-square tests to examine relationships between categorical variables.
- Regression or classification models to investigate purchasing patterns.

These are proposed extensions; the Chapter 1 exercise itself concentrates on identifying variable types.

Main lesson

Statistics helps companies understand customer differences and design more targeted marketing strategies rather than advertising indiscriminately.

28. Case 5: Clear Mountain State University surveys

Textbook page 59

Background

The Student News Service at Clear Mountain State University conducts an undergraduate survey containing 14 questions.

It receives 62 undergraduate responses.

The dean of students conducts a similar graduate survey and receives 44 responses.

The chapter asks students to examine the questionnaires and classify the variables as categorical, discrete numerical, or continuous numerical.

The actual questionnaires and datasets are separate downloadable materials and are not reproduced in the uploaded chapter.

Classroom application

You can extend this exercise into a discussion about student satisfaction and university management.

Possible variables include:

- Academic program.
- Study hours.
- Student satisfaction.
- Campus service usage.
- Travel time to campus.
- Employment status.

These are suggested variables rather than confirmed contents of the original questionnaires.

Questions to ask

- Are 62 undergraduate responses enough to represent all undergraduates?
- Were students randomly selected?
- Could nonresponse bias affect the results?
- Are undergraduate and graduate students meaningfully different populations?
- Should the university use stratified sampling to ensure adequate representation from both groups?

Main lesson

Statistics is relevant not only to commercial organizations but also to educational administration and institutional decision-making.

29. Additional applied examples in the chapter

Beyond the main cases, the chapter includes exercises involving:

| Application | Statistical concept |

| ---------- | ---------- |

| Initial public offerings in the UK | Population versus sample |

| Hotel room categories | Ordinal measurement |

| Restaurant delivery zones | Nominal measurement |

| Mortgage applications | Variable classification |

| Retail shopping behavior | Survey design |

| Employee motivation by department | Stratified sampling |

| Sales invoice auditing | Simple random and systematic sampling |

| College satisfaction surveys | Sampling design |

| Mobile shopping surveys | Survey errors |

| Supply-chain executive surveys | Survey credibility |

| Consumer media behavior | Survey evaluation |

| Retirement-planning awareness | Population and sample identification |

| Cat food purchasing behavior | Variable types and questionnaire design |

These smaller exercises are valuable because they expose students to different industries and show that the same statistical principles apply across business functions.

Part IX — Bringing the Chapter to Life in Class

30. Suggested teaching sequence

For a master's-level Business Statistics class, I would organize the chapter around business decisions rather than presenting definitions in isolation.

Suggested 90-minute session

0–15 min

Start with the GT&M acquisition

Ask what information is needed to recommend a purchase price.

15–30 min

Define variables and measurement scales

Use sales, profitability, satisfaction, and customer segmentation examples.

30–45 min

Discuss data collection and quality

Compare surveys, transactions, experiments, and secondary sources.

45–65 min

Compare sampling methods

Use university students, bank branches, and factory production examples.

65–80 min

Explore survey errors

Present the Literary Digest case and a flawed student satisfaction survey.

80–90 min

Close with a business decision

Use CardioGood Fitness or Ashland MultiComm to connect the concepts.

31. Five questions to stimulate discussion

These questions can be used throughout the lecture.

Question 1 — Business acquisition

A company reports that revenue increased by 30%. Does this necessarily mean the company is financially healthier?

Students should recognize that revenue growth alone is insufficient. Profitability, expenses, cash flow, and other factors matter.

Question 2 — Sampling

Would you prefer a voluntary survey of 100,000 customers or a properly selected random sample of 1,000 customers?

The answer depends on the research objective and design, but the random sample may provide more defensible population estimates despite being much smaller.

Question 3 — Customer satisfaction

A bank reports 95% satisfaction based on responses from its mobile app users. Can it conclude that 95% of all bank customers are satisfied?

Not necessarily. The survey may exclude branch-only customers and may suffer from nonresponse bias.

Question 4 — Marketing

CardioGood Fitness wants to identify the characteristics of customers who purchase premium treadmills. Which variables should be collected?

Students can propose income, age, fitness level, expected usage, and purchasing behavior.

Question 5 — Business performance

A cable television provider is losing subscribers. Does this prove that its prices are too high?

No. Several explanations are possible, including competition, streaming alternatives, service quality, and changing customer preferences.

32. Short knowledge check

Test your understanding

0 of 6 answered

1\. A university randomly selects students from every academic program to ensure representation.

Cluster

Stratified

Systematic

Convenience

2\. A company randomly selects 10 branches and surveys all employees in those branches.

Cluster

Stratified

Simple random

Systematic

3\. An auditor selects every 50th invoice after a random starting point.

Stratified

Judgment

Systematic

Cluster

4\. A survey of all bank customers excludes those who do not use the mobile app.

Sampling error

Measurement error

Nonresponse error

Coverage error

5\. A random sample estimates satisfaction at 72% when the true population value is 70%.

Sampling error

Coverage error

Nonresponse error

Measurement error

6\. A questionnaire asks customers how much they like the company's excellent service.

Coverage error

Measurement error

Sampling error

Cluster error

Part X — Final Chapter Synthesis

33. What students should remember

The chapter's most important lessons can be summarized as follows.

1\. Define the business problem before collecting data.

Statistical work should begin with a clear decision or research objective.

2\. Define variables precisely.

Ambiguous variables lead to inconsistent measurements and unreliable conclusions.

3\. Understand data types and measurement scales.

They determine which comparisons and statistical operations are meaningful.

4\. Select data sources carefully.

Primary and secondary data, experiments, surveys, observations, and operational records serve different purposes.

5\. Distinguish populations from samples.

Sample statistics are used to estimate population parameters, subject to sampling uncertainty and other possible errors.

6\. Prepare data before analyzing it.

Missing values, outliers, inconsistent formats, and incorrect categories can undermine analysis.

7\. Choose sampling methods based on the research objective.

Simple random, systematic, stratified, and cluster sampling have different strengths and limitations.

8\. Understand the difference between stratified and cluster sampling.

Stratified sampling selects individuals from every important subgroup. Cluster sampling selects some groups, usually to improve practicality and reduce costs.

9\. Recognize the four survey errors.

Coverage, nonresponse, sampling, and measurement errors can all affect survey validity.

10\. Large datasets do not automatically produce reliable conclusions.

The Literary Digest case demonstrates why sampling design and data quality matter more than sample size alone.

11\. Evaluate business claims critically.

The Whitney Wireless digital case illustrates how ambiguous definitions and misleading visualizations can create unsupported conclusions.

12\. Statistics is part of business decision-making, not merely mathematical computation.

The cases demonstrate applications in acquisitions, marketing, telecommunications, university administration, customer research, and quality management.

34. Overall assessment of the chapter

Chapter 1 is primarily a methodological foundation rather than a computational statistics chapter.

It introduces relatively few calculations, focusing instead on the decisions researchers must make before formal analysis.

Its strongest teaching contributions are:

- Translating business objectives into measurable variables.
- Understanding the structure and quality of data.
- Selecting appropriate sampling designs.
- Recognizing sources of survey bias and uncertainty.
- Critically evaluating statistical claims.

For a master's-level course, I would place particular emphasis on the GT&M acquisition, CardioGood Fitness customer segmentation, Ashland MultiComm's declining subscriptions, the Literary Digest polling failure, and the Whitney Wireless misleading visualization case.

Together, these cases demonstrate that statistics can support financial decisions, marketing strategy, operational improvements, and the evaluation of business evidence.

The chapter's central principle is worth repeating:

Statistical methods cannot rescue a fundamentally flawed data collection process.

And the broader message for your students is:

Statistics gives managers a disciplined way to turn business questions into evidence, evaluate uncertainty, and make better-informed decisions.